01 / PROBLEM AND CONTRIBUTION

Differential transconductance amplifier

EEE 433/591 · Analog IC Design · Lab 4 | Fall 2025

RESULT AT A GLANCE

Evaluated a differential-to-single-ended OTA with reported AC gain of 33.55 dB, 224.25 kHz bandwidth, and 10.69 MHz unity-gain frequency.

Engineering question

How can a transistor-biased differential pair provide gain while rejecting common-mode and supply disturbances?

Situation

A differential OTA must amplify the intended input difference while controlling its operating point and response to unwanted inputs. This lab examines a transistor tail source, active load, DC feedback, and separate AC/transient test configurations.

Task

Design and evaluate the differential-to-single-ended OTA, including differential gain, common-mode response, supply response, and a feedback step test.

My contribution

Documented device sizing and operating points, differential and common-mode simulations, supply-response plots, and unity-feedback transient behavior.

Tools and concepts

Operational transconductance amplifiers / Differential pairs / Cadence Virtuoso / AC analysis / Common-mode analysis

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation

Established transistor bias and the DC feedback testbench, and tabulated device sizes and branch currents in the report.

02 / Implementation

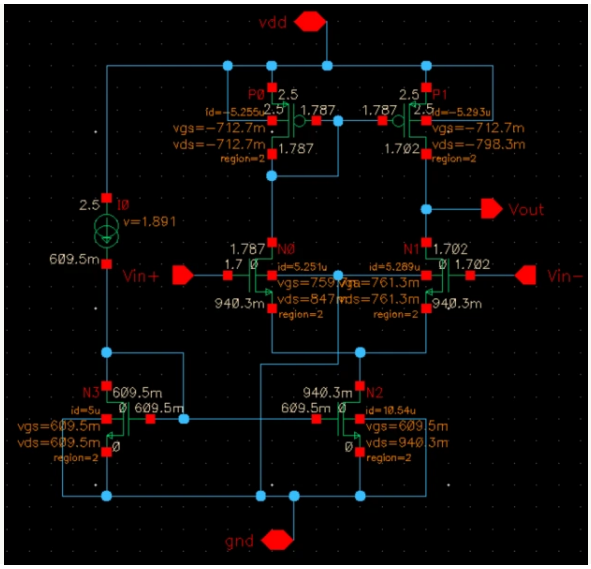
Measured the differential AC response. The report lists 33.551 dB low-frequency gain, 224.254 kHz bandwidth, and 10.6883 MHz unity-gain frequency.

03 / Implementation

Examined common-mode and supply-response plots separately from the differential test, then simulated a step with unity-gain feedback. Kept these test configurations distinct when interpreting the results.

Separate signal paths

Differential gain, common-mode gain, and supply coupling describe different inputs to the amplifier. Their test configurations must be preserved when forming rejection ratios or comparing frequency responses.



OTA transistor operating point

Source report p. 5: differential pair, active load, and transistor bias network.

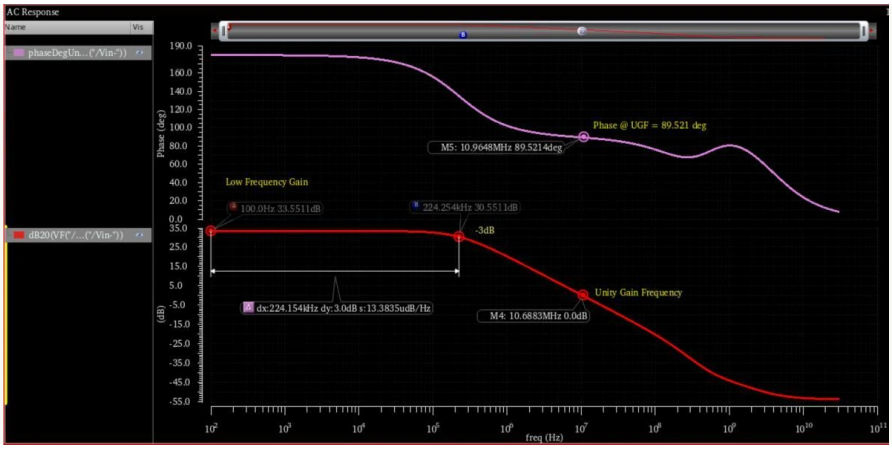
03 / RESULTS AND PERSPECTIVE

Evidence and engineering perspective

AC gain	33.55 dB
-3 dB bandwidth	224.25 kHz
Unity-gain frequency	10.69 MHz

Reported result

The differential AC plot supports approximately 33.55 dB gain, 224.25 kHz bandwidth, and 10.69 MHz unity-gain frequency. Common-mode, supply-rejection, and step-response figures broaden the study beyond gain alone.



Differential AC response

Source report p. 7: low-frequency gain, bandwidth, and unity-gain markers.

Scope and limitations

Phase at unity gain is not sufficient by itself to establish loop phase margin. The report has inconsistent transient amplitude, CMRR arithmetic, and supply-noise wording; those quantities are not promoted as verified headline metrics. The copied F2021 template heading is not used as the project date.

Possible next steps - proposed work

- Reconcile CMRR and supply-noise calculations with consistently defined transfer functions.
- Run a documented loop-gain measurement before quoting a phase margin.

Source material: EEE 433/591 Lab 4 submitted report, physical PDF pp. 5-7, 9-16; differential AC markers on p. 7. Semester follows the supplied Semester 1 course context.